

Predicting Medical Costs and Motivation Dimension Reduction: An Application of Optimal Scaling Methods

Exploratory Data Analysis Assignment 2

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Introduction

The United States operates through a combination of private health insurance, employer-sponsored plans, and public programs such as Medicare and Medicaid, which together aim to provide access to medical care for diverse populations (ISPOR, n.d.). However, the decentralised nature of the system results in wide variations in healthcare coverage, service costs, and spending patterns across individuals and regions (Mandel, 2023). These differences make the prediction and understanding of medical expenses an important issue for both policymakers and insurers. Thus, in the first part of the assignment, we analyse a subset from the Medical Cost Personal Dataset (Lantz, 2019) to explore which personal characteristics predict individual medical insurance charges. The dataset includes information on age, sex, body mass index (BMI), smoking status, residential region, and individual medical costs billed by health insurance. The objective of this analysis is to predict an individual's medical charges based on these explanatory variables using optimal scaling regression in SPSS (IBM Corp., 2024). Through this analysis, we aim to identify which factors most strongly contribute to higher healthcare costs and assess how well the model predicts insurance charges.

In the Netherlands, the unemployment rate in June 2025 was 3.8%, equivalent to approximately 386,000 individuals, and rose slightly to 3.9% later in the year, remaining low by historical standards (CBS, 2025). Despite this, a considerable portion of the population is not active in the labour force, and many unemployed individuals receive government benefits while seeking to re-enter employment. Understanding motivational factors that influence readiness to re-enter the labour market is therefore of practical and policy relevance. Accordingly, the second part of this assignment analyses ten questionnaire items designed to measure motivation and work readiness among (partially) unemployed benefit recipients. Using optimal scaling principal components analysis (CATPCA) in SPSS (IBM Corp., 2024), the analysis examines the latent structure underlying these items, assesses whether work readiness is best conceptualised as a unidimensional or multidimensional construct, and evaluates whether optimal scaling can reduce information loss when extracting interpretable components that capture individuals' motivational readiness for employment within the Dutch labour market.

Motivation to re-enter the labour market is a complex construct typically measured using ordinal questionnaire items, for which standard linear methods may be suboptimal. Therefore, the second part of this assignment applies Categorical Principal Components Analysis (CATPCA) with optimal scaling to investigate the latent structure underlying ten work-readiness items. CATPCA is particularly suitable in this context because it allows ordinal variables to be transformed optimally without imposing unjustified interval-level assumptions, while simultaneously reducing dimensionality and minimising information loss. The goal is to assess whether work readiness is best conceptualised as a unidimensional or multidimensional construct and to evaluate the interpretability and quality of the resulting component structure.

Methods

Optimal Scaling Regression

The analysis strategy followed a stepwise approach in which the level of measurement restrictions was systematically varied using *Optimal Scaling Regression*. The strategy began with the most restrictive model and subsequently relaxed these constraints to examine predictor behaviour under increasing flexibility, before reintroducing selected restrictions to improve interpretability and model parsimony. Model performance was evaluated using both the Apparent Prediction Error (APE) and the Expected Prediction Error (EPE), the latter estimated via the .632 *bootstrap* method with 1,000 resamples. The APE provides an in-sample estimate of fit, reflecting how well the model predicts the data used to train it. In contrast, the EPE offers a more realistic measure of out-of-sample predictive performance by repeatedly resampling the data and correcting for overfitting.

In the initial analysis (Model 1), Optimal Scaling Regression was fitted using the default discretisation settings. Since all predictors were integer-valued, they were left unchanged. The dependent variable, charges, contains zero values, which are treated as missing in SPSS. Therefore, a *Grouping* discretisation method was applied, dividing the outcome into seven approximately normally distributed categories. As the variable has exactly seven observed categories, this transformation does not result in a loss of information. All predictors were specified at a numeric scaling level, imposing linear relationships with the outcome and preserving equal distances between scale points. The only exception was “region”, which was treated as nominal, allowing its four categories (northeast, southeast, southwest, northwest) to be modelled as qualitatively distinct without assuming an inherent order.

Model 2 relaxed these restrictions by allowing all predictors to be optimally transformed at a nominal scaling level, while the dependent variable remained numeric with *Grouping* discretisation into seven categories. Discretisation methods were adapted to the type of each predictor. The binary variables “sex” and “smoker” were discretised using *Grouping* into two categories, while “region” was grouped into four categories. The continuous predictors “age” and “bmi” were discretised using the *Multiplying* method, which standardises the original values, multiplies them by a constant, rounds them, and rescales them so that the minimum value equals one. This model allowed the predictors to vary freely, without imposing linearity or ordering constraints.

In Model 3, restrictions were partially reintroduced to improve interpretability while retaining sufficient flexibility. The dependent variable “charges” was again specified as numeric, with *Grouping* discretisation into seven categories. The binary predictors “sex” and “smoker” were set to numeric scaling, as their dichotomous nature yields identical results across alternative scaling options while offering a more parsimonious and interpretable representation. “Region” was retained as nominal, reflecting the absence of a meaningful ordering among its categories.

Finally, the continuous predictors “age” and “bmi” were modelled using spline-nominal transformations (degree 2 with six internal knots), allowing for nonlinear relationships with the outcome while preserving smoothness.

Optimal Scaling Dimension Reduction

Data Preparation

The dataset consisted of 10 Likert-scale items measuring aspects of work-readiness ($N=400$). Missingness was low (13 missing values across a subset of variables), and the assignment instructed removal of incomplete cases, so all analyses were conducted on complete observations. Several variables displayed categories with very low frequencies (e.g., only two responses in category 5 for one item). This issue was expected to result in unstable optimal-scaling transformations, which were assessed using bootstrap confidence intervals.

Optimal Scaling and PCA

The goal was to determine whether the construct “Work Readiness” is uni- or multidimensional and to assess whether optimal-scaling PCA can reduce information loss relative to standard PCA. Three analytical decisions guided the process: (1) the discretisation level, (2) the level of optimal scaling, and (3) the number of dimensions to retain.

Choice of Discretisation

Because all items used a 5-point Likert scale and the total number of categories per variable was below the threshold requiring further categorisation, no additional binning was applied. Following the assignment instructions, items were ranked prior to analysis to ensure a consistent ordering of categories. It is important to note that ranking itself is not an optimal-scaling transformation, but a preparatory step. The actual transformations were applied during CATPCA by specifying the appropriate scaling level (nominal or ordinal), which determines how category quantifications are optimally estimated.

Choice of Scaling Level

Analyses proceeded from less restrictive to more restrictive scaling: first treating items as nominal, then as ordinal. Interval scaling was not explored further because equal spacing between Likert responses cannot be assumed, and spline transformations are not recommended for fewer than seven categories.

Results

Optimal Scaling Regression

The model summary of our initial analysis (Model 1) achieved an R^2 of .744, suggesting that approximately 74.4% of the variance in the medical charges is explained by the predictors. After adjusting for the number of predictors, the adjusted $R^2 = .740$ confirmed the explanatory power of the current model. The Apparent Prediction Error (APE) of .256 indicated that approximately 25.6% of the variance in the outcome remains unexplained by the model. The Expected Prediction Error (EPE) of .269 (SE = .27) suggested that around 26.9% of variance would remain unexplained when generalising to a new sample. Model 2 achieved an R^2 of .787, meaning that approximately 78.7% of the variance in the medical charges was explained by the predictors. After adjusting for the number of predictors, the adjusted R^2 was .742. The APE decreased to .213, suggesting better in-sample predictive accuracy. This improvement in in-sample performance is consistent with the increased flexibility of the model. The EPE increased to .323 (SE = .033), indicating better performance when generalising to new samples. Model 3 yielded an R^2 of .758, indicating that 75.8% of the variance in medical charges was explained by the predictors. After adjusting for the number of predictors, the adjusted R^2 was 0.745. The APE and EPE achieved by the model were .242 and .273, respectively. All model estimates are summarised in Table 1.

Taken together, Model 3 provides the best balance between model fit and predictive performance. While Model 2 yields superior in-sample accuracy (APA = .213), its substantially higher EPE (.323) indicates overfitting. Model 3 achieves more stable out-of-sample performance (EPE = .273) while retaining adequate explanatory power ($R^2 = .787$), and can therefore be regarded as the preferred model.

Table 1

Model Fit Statistics for the Optimal Scaling Regression Predicting Individual Medical Costs

Model	R^2	Adj-R^2	APE	EPE
<i>Model 1</i>	.744	.740	.256	.269
<i>Model 2</i>	.787	.742	.213	.323
<i>Model 3</i>	.758	.745	.242	.273

Note. R^2 and Adjusted R^2 reflect the proportion of variance in medical costs explained by each model. APE (Apparent Prediction Error) represents in-sample prediction error, whereas EPE

(Expected Prediction Error) is estimated using the .632 bootstrap with 1,000 resamples to provide an out-of-sample performance estimate. All values are based on standardised variables.

In this model, the standardised coefficients (β) showed the relative contribution of each predictor to explaining variation in individual medical costs after optimal scaling. As shown in Table 2, smoking status is by far the strongest predictor ($\beta = .778, p < .001$), indicating that smokers incur substantially higher medical costs than non-smokers, even after controlling for age, sex, BMI, and residential area. The age of the primary beneficiary also shows a strong positive association with medical charges ($\beta = .320, p < .001$), suggesting that older individuals tend to have higher insurance costs. Body mass index (BMI) is likewise a highly significant predictor ($\beta = .187, p < .001$), meaning that higher BMI values are linked to increased medical expenses. In contrast, sex ($\beta = .020, p = .458$) and residential area ($\beta = .026, p = .169$) are not statistically significant, suggesting minimal differences in medical costs across these categories.

Table 2

Standardised Coefficients for the Optimal Scaling Regression Predicting Individual Medical Costs (Model 3)

Predictor	β	SE	F	p
Age of primary beneficiary	.320	.027	141.88	< .001
Sex	.020	.027	.552	.458
Body mass index (BMI)	.187	.026	51.32	< .001
Smoking status	.778	.024	1080.04	< .001
Residential area	.026	.020	1.691	.169

Note. Dependent variable = individual medical costs billed by health insurance.
 β = standardised coefficient; SE = bootstrap standard error (1,000 resamples).

The importance values showed that smoking status contributes most strongly to the prediction of medical costs (importance = .803), followed by age (.128) and BMI (.063),

reflecting their relative influence in the model. Sex (.002) and residential area (.003) contribute very little, consistent with their nonsignificant regression coefficients. The tolerance values, both before and after transformation, are all high ($>.95$), indicating that none of the predictors exhibit problematic multicollinearity. This suggests that the predictors provide largely independent information and that the model's estimates are stable.

Table 3

Correlations, Importance Values, and Tolerance Statistics for Predictors of Individual Medical Costs (Model 3)

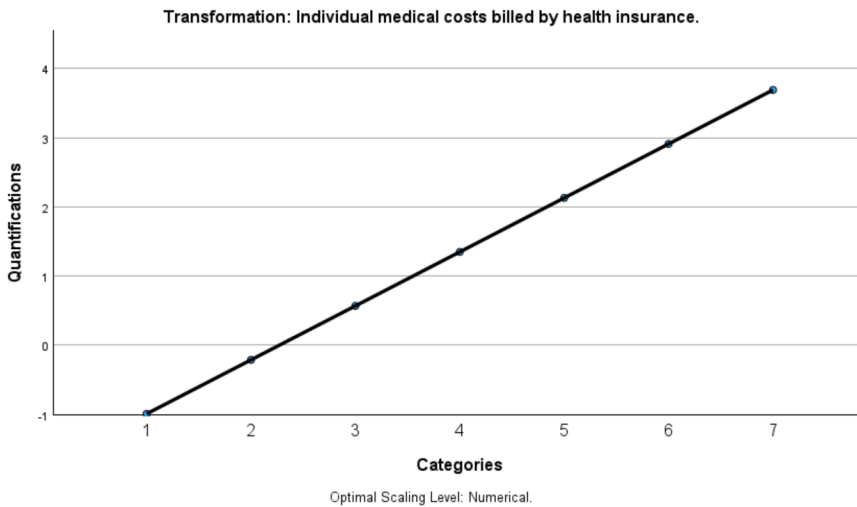
Predictor	Importance	Tolerance (after)	Tolerance (before)
Age of primary beneficiary	.128	.997	.992
Sex	.002	.990	.988
Body mass index (BMI)	.063	.963	.957
Smoking status	.803	.985	.987
Residential area	.003	.968	.954

Note. Dependent variable = individual medical costs billed by health insurance. Importance values indicate each predictor's relative contribution to the model. Tolerance values represent the degree of multicollinearity; values closer to 1 indicate minimal overlap among predictors.

The transformation plots showed that “charges” increase linearly across their seven categories, indicating a strong monotonic relationship with the transformed scale (Figure 1). In contrast, the “age” transformation reveals a nonlinear association with “charges” (Figure 2). Predicted charges increase gradually across lower age categories, followed by a steeper rise in mid to later age categories. The relationship appears to level off briefly in the middle range before increasing more sharply toward higher age categories. At the upper extreme, a decline in the transformation is observed. Overall, the transformation indicates that the effect of age on medical charges is not constant across the age range, supporting the use of a flexible, nonlinear specification for this predictor.

Figure 1

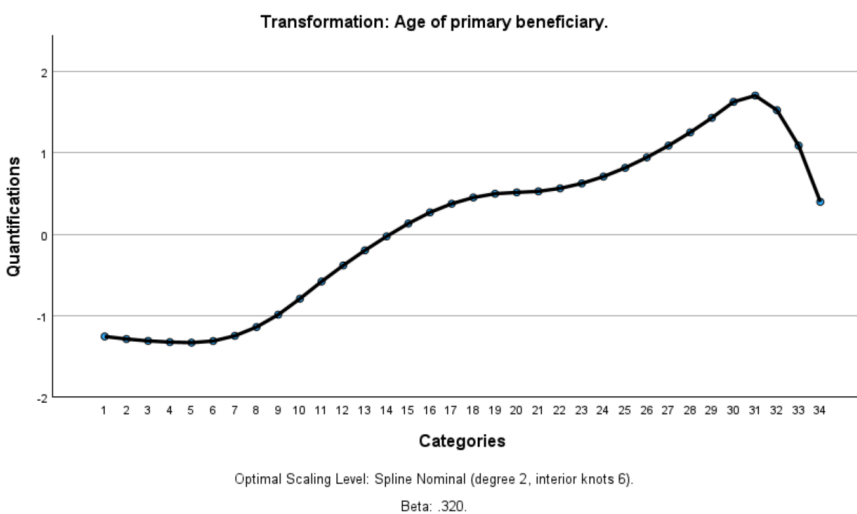
Optimal scaling transformation of medical charges.



Note. Quantifications increase in a nearly perfect linear pattern across the seven charge categories, indicating a strong monotonic relationship consistent with numerical scaling.

Figure 2

Optimal scaling transformation of age.



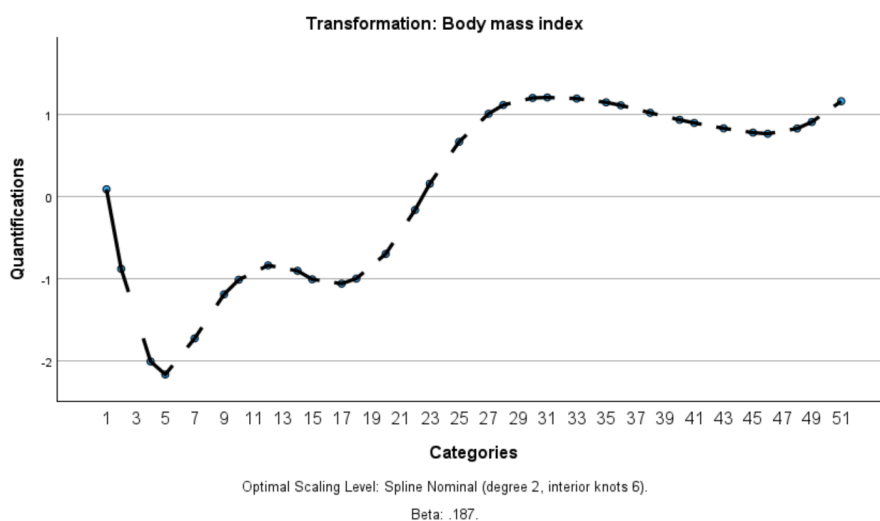
Note. The figure shows the spline-nominal transformation (degree 2, 6 knots) of age obtained from the optimal scaling regression. “Age” was discretised using the Multiplying method and

therefore represents standardised age categories rather than age in years. The x-axis indexes ordered age categories, while the y-axis displays the corresponding optimal quantifications.

In Figure 3, the spline-nominal transformation of standardised body mass index (BMI) indicates a strong nonlinear association with medical charges. Lower BMI categories are associated with substantially lower predicted charges, followed by a steep increase across the mid-range categories. The relationship appears to plateau at higher BMI categories, with predicted charges remaining relatively stable across much of the upper range. A slight increase is observed at the extreme upper end of the distribution, though this pattern should be interpreted cautiously, as it may reflect increased uncertainty due to fewer observations in these categories. Overall, the transformation suggests that the effect of BMI on medical charges is strongest at lower to moderate values and diminishes at higher levels, supporting the use of a flexible nonlinear specification.

Figure 3

Optimal scaling transformation of BMI



Note. The figure shows the spline-nominal transformation (degree 2 with six interior knots) of body mass index obtained from the optimal scaling regression. BMI was discretised using the Multiplying method and therefore represents standardised BMI categories rather than raw BMI values. The x-axis indexes ordered BMI categories, while the y-axis displays the corresponding optimal quantifications.

Optimal Scaling Dimension Reduction

a) Nominal Scaling (Unrestricted)

A two-dimensional OS-PCA was performed using nominal scaling. Variance accounted for by the two-dimensional solution ranged between 58.2% and 63.8% across components. Loadings showed that items clustered differentially across the two dimensions, although the interpretability of the transformed variables was limited. Bootstrap confidence intervals revealed instability for variables with sparse categories. One item in particular (“I don’t think I’m able to search for a new job”) showed poor representation, wide CIs, and unusually high residuals, indicating misfit.

b) Ordinal Scaling (More Restrictive)

Using ordinal scaling yielded VAF values between 47.5% and 63.3%, with a mean similar to nominal scaling ($\approx 60\%$). While VAF did not improve, the transformations were far more interpretable, preserving the expected ordering without forcing equal spacing. Bootstrap CIs were narrower for most items relative to the nominal case, though the same problematic item continued to exhibit instability. Residual plots showed substantial improvement, with most variables aligning well to the diagonal, except for the outlier.

Given the conceptual appropriateness of ordinal scaling and the improved interpretability and residual structure, ordinal scaling was selected for subsequent analyses.

Importantly, the use of ordinal scaling did not result in a meaningful loss of model fit relative to nominal scaling, as the variance accounted for remained comparable across both approaches. Given this equivalence in goodness-of-fit, ordinal scaling was preferred because it preserves the expected ordering of Likert-scale responses while yielding more interpretable transformations and more stable bootstrap confidence intervals.

Determination of Dimensionality

A scree plot on the full 10-dimensional solution was used to identify an elbow. Follow-up scree analyses on 1, 2, and 3 retained dimensions consistently suggested an optimal solution of two dimensions. These findings were replicated when applying both unrotated and rotated solutions.

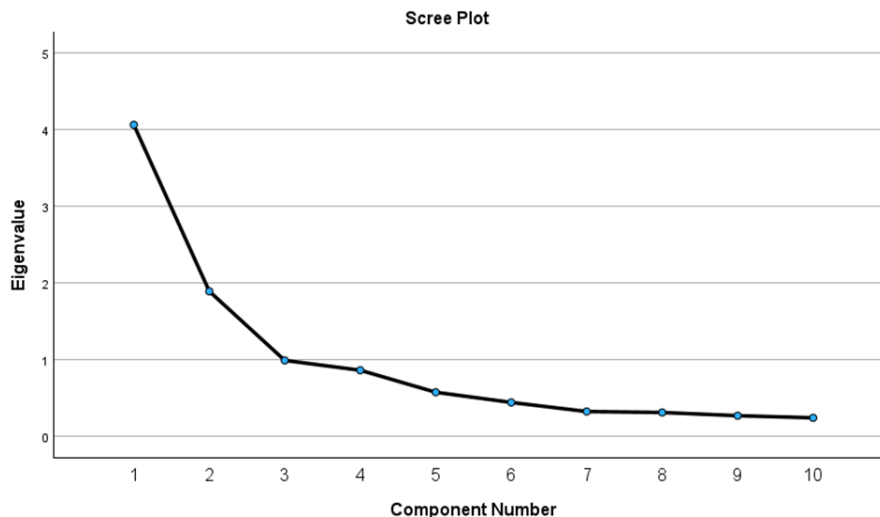
Rotation and Interpretation

A two-dimensional varimax-rotated ordinal CATPCA solution was retained. Rotation substantially clarified the component structure, and removing the problematic item improved the overall configuration by yielding more coherent and interpretable loading patterns. The optimally scaled PCA indicated that the construct of work readiness is multidimensional rather than unidimensional. Two dimensions captured the meaningful structure of the remaining nine items, while one item (“I don’t think I am able to search for a new job”) consistently failed to load

clearly on either dimension and exhibited unstable bootstrap confidence intervals. This item was therefore excluded from substantive interpretation.

Figure 4

Scree plot of eigenvalues for the CATPCA solution.



Note. The figure shows the scree plot of eigenvalues obtained from the full CATPCA solution including all ten work-readiness items. The plot displays a clear elbow after the second component, indicating diminishing gains in explained variance beyond two dimensions. Based on the elbow criterion, a two-dimensional solution was retained for further analysis. Similar patterns were observed for both unrotated and rotated solutions.

Interpretation of the rotated solution focused on the magnitude and direction of item loadings rather than on grouping items into clusters. Items reflecting precontemplation (Motivation 3, 4, and 5) loaded strongly and positively on Dimension 1, whereas items reflecting contemplation (Motivation 1, 2, and 8) loaded strongly but negatively on the same dimension. Importantly, a negative loading indicates a strong association with the component in the opposite direction, rather than a weak or absent relationship. Thus, both contemplation and precontemplation items jointly define Dimension 1 as a single continuum.

Dimension 1 was therefore interpreted as **self-awareness / perceived need for change**, with higher positive scores reflecting denial or low perceived need for change and higher negative scores reflecting greater awareness and recognition of the need for behavioural change.

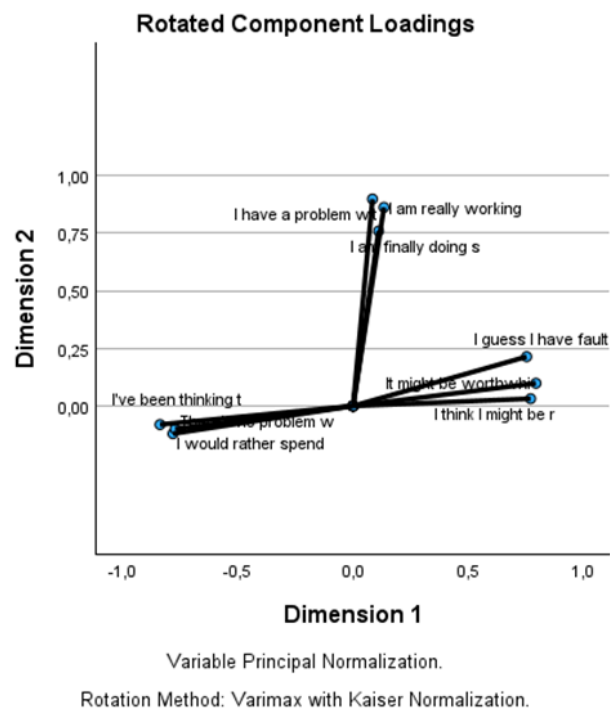
In applied contexts, this dimension could be operationalised as a single scale by reverse-coding the precontemplation items prior to scale construction.

Items reflecting action-oriented behaviour (Motivation 6, 9, and 10) loaded highly on Dimension 2 and showed minimal association with Dimension 1. These items capture concrete behavioural engagement and active efforts to improve one's employment situation. Dimension 2 was therefore interpreted as **action orientation / behavioural engagement**.

The use of optimal scaling did not substantially increase the variance accounted for compared to numeric scaling, but it improved the interpretability of the component structure and the stability of the transformations, particularly under ordinal scaling. The persistent instability of the excluded item across scaling levels suggests a limitation in content validity rather than an issue attributable to the transformation method.

Figure 5

Component loadings plot for the two-dimensional ordinal CATPCA solution.



Note. The figure displays the component loadings of the nine retained work-readiness items on the two-dimensional ordinal CATPCA solution after varimax rotation. Dimension 1 represents *self-awareness / perceived need for change*, while Dimension 2 reflects *action orientation /*

behavioural engagement. Items are positioned according to their loadings on both dimensions, illustrating the emergence of distinct motivational patterns. One unstable item was excluded prior to interpretation due to poor representation and wide bootstrap confidence intervals.

Discussion

Optimal Scaling Regression

The aim of this analysis was to predict an individual's medical costs billed by health insurance using demographic and health-related variables. Across the optimal scaling regression models tested, Model 3 offered the best balance of explanatory power and predictive accuracy, explaining 76% of the variance in medical charges. The findings consistently showed that smoking status was the strongest predictor of higher medical costs, followed by age and BMI, whereas sex and residential region contributed very little to the prediction. The transformation plots further indicated nonlinear relationships for age and BMI, highlighting that medical costs rise steeply in mid-to-late adulthood and at both moderate and very high BMI levels. A major strength of this approach is the use of optimal scaling, which flexibly models nonlinear relationships while minimising multicollinearity, as reflected in the high tolerance values.

Although the model performed well, several limitations should be noted. The sample includes only a small set of predictors and reflects a single, cross-sectional U.S. insurance context, limiting both the completeness of the model and the generalisability of the findings to other populations or healthcare systems. In terms of methodology, optimal scaling regression comes with some limitations as well. Nonlinear relationships are restricted by the chosen transformation type (spline, ordinal, and numeric transformations), meaning the model does not freely estimate the most appropriate functional form, but it is imposed by the researcher. In addition, the resulting quantifications replace the original variable units, making transformed predictors harder to interpret. CATREG also cannot model interactions or more complex dependency structures, which limits its ability to capture realistic patterns. Finally, all optimal scaling quantifications are sample-dependent and may not generalise well to new populations.

Future research could extend these findings by incorporating additional predictors that capture medical risk more comprehensively, such as chronic health conditions and interaction effects. It would also be valuable to compare optimal scaling regression with more flexible machine-learning approaches (e.g., random forest models) to assess whether nonparametric methods better capture complex nonlinearities in medical cost data.

Optimal Scaling Dimension Reduction

The results of the optimal scaling PCA indicate that the set of motivation items does not reflect a single, unidimensional construct of work readiness. Instead, the analysis consistently

supported a two-dimensional structure, which proved robust across different scaling assumptions and rotation choices. This finding suggests that work readiness, as operationalised by the present items, comprises at least two distinct but related motivational components rather than a single underlying trait.

The first dimension captured variation in self-awareness and perceived need for change, with contemplation and precontemplation items loading strongly in opposite directions. Importantly, these items did not represent separate latent constructs, but rather defined a single continuum, ranging from denial or low perceived need for change to increased awareness and recognition of the need for behavioural change. The second dimension reflected action orientation and behavioural engagement, capturing concrete efforts and intentions to actively improve one's employment situation. This dimensional structure aligns conceptually with stage-based models of behavioural change, which distinguish between awareness-related and action-oriented components of motivation.

CATPCA provided clear advantages in terms of interpretability and stability. Ordinal scaling preserved the ordered nature of Likert-scale responses without imposing equal spacing assumptions, resulting in transformations that were more conceptually meaningful. Furthermore, bootstrap confidence intervals revealed improved stability for most items under ordinal scaling, supporting the appropriateness of this approach for ordinal questionnaire data. These findings illustrate that the primary benefit of optimal scaling in this context lies not in maximising explained variance, but in producing interpretable and theoretically coherent component structures.

At the same time, the analysis highlights important limitations. One item consistently exhibited unstable quantifications and failed to load clearly on either dimension, even after changes in scaling level and rotation. This persistent instability suggests a problem with the content validity of the item rather than with the transformation method itself. The need to exclude this item underscores the sensitivity of CATPCA to poorly functioning indicators and illustrates how optimal scaling can serve as a diagnostic tool for identifying weaknesses in measurement instruments. Nevertheless, the exclusion of an item and the modest proportion of variance explained indicate that the current set of items only partially captures the complexity of work readiness.

Regarding solution quality, the final two-dimensional ordinal CATPCA solution demonstrated acceptable stability and interpretability for the majority of items. Bootstrap analyses indicated that most category quantifications were reasonably stable, lending support to the robustness of the extracted dimensions. However, the relatively moderate variance accounted for, combined with the dependence of optimal scaling transformations on the specific sample, cautions against overgeneralisation. The resulting dimensions should therefore be interpreted as exploratory rather than definitive representations of the work-readiness construct.

In practical terms, the findings suggest that work readiness may be meaningfully operationalised using two separate but related scales: one reflecting self-awareness or perceived need for change (potentially constructed by reverse-coding precontemplation items), and another capturing action-oriented motivation. For future research, scale refinement through the development of additional, better-targeted items would likely improve both reliability and construct coverage. Replication in independent samples would also be necessary to assess the generalisability of the dimensional structure and the stability of the optimal scaling transformations.

In conclusion, optimal scaling PCA proved to be a valuable exploratory tool for analysing ordinal motivation data. While it did not substantially increase explained variance relative to standard PCA, it enhanced interpretability, facilitated the detection of problematic items, and yielded a theoretically coherent multidimensional representation of work readiness. These results support the use of CATPCA as a method for exploratory scale evaluation and dimensionality assessment when dealing with ordinal questionnaire data.

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Peer adjustment

Alexandra Constantin (++): Authored the Introduction and Methods section corresponding to Part 1 of the assignment. Performed Analysis corresponding to Part 1 of the assignment, authored the corresponding Results section and implemented the feedback received. Authored the Discussion and Conclusions corresponding to Part 1 of the assignment. Coordinated the group work and offered support in critically analyzing the remaining sections.

Eulàlia Gómez-Aguiló (+): Co-authored Introduction (included details regarding Part 2 of the assignment). Authored Methods section for the analysis corresponding to Part 2 of the assignment. Performed Analysis corresponding to Part 2 of the assignment, authored the corresponding Results section and implemented the feedback received. Authored the Discussion and Conclusions corresponding to Part 2 of the assignment.

Taymaz Ghaneh (0): Reviewed texts in-between deadlines. Offered theoretical background explanations and critically reviewed the Analysis and Results. Supported and co-authored Analysis and corresponding Results sections.

Use of AI

AI-based tools were used to proofread the report and troubleshoot SPSS syntax errors.